



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
FIRST SESSIONAL
SUBJECT: (CE-514) ADVANCED ALGORITHMS

Examination : B.Tech Semester-V
Date : 03/08/2022
Time : 02:30 P.M. - 03:45 P.M.

Seat No : 32
Day : Wednesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

CO2 (a) How many augmented paths will be calculated in the worst case of Ford Fulkerson's algorithm to find Max Flow in the graph shown in the figure 1 (A)?

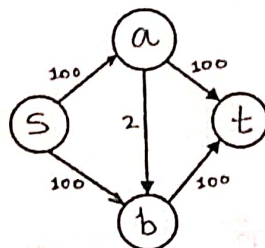


Figure 1 (A)

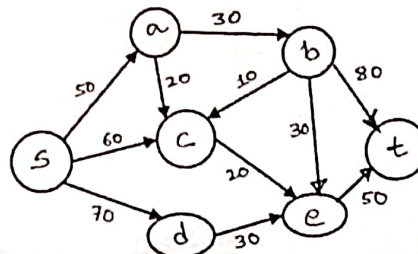


Figure 1 (B)

CO1 (b) Find the Min S-T Cut in the graph shown in the figure 1 (B). [2]

CO1 (c) State true or false and justify your answer: Ford Fulkerson's algorithm finds Max Flow in polynomial time. [2]

CO1 (d) Differentiate: Las Vegas and Monte Carlo types of Randomized algorithms. [2]

CO2 (e) Consider the experiment of rolling a fair die 100 times. Say, X is the Random variable which indicates the total number of times an even number appeared. Find out $E[X]$ using Indicator Random Variables. [2]

CO1 (f) Let $A[1..n]$ be an array of n distinct numbers. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an inversion of A. Suppose that the elements of A form a uniform random permutation of $\{1, 2, \dots, n\}$. Use indicator random variables to compute the expected number of inversions. [2]

Q.2 Attempt Any Two of the following questions.

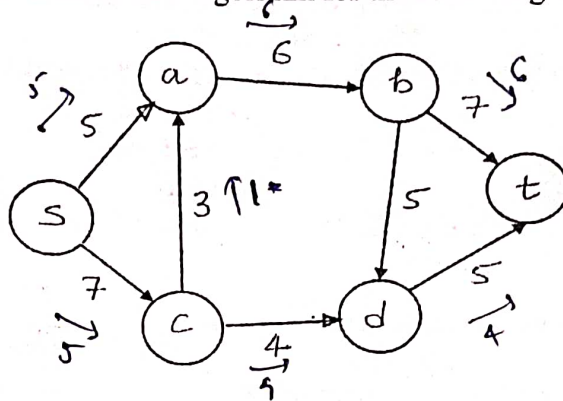
CO1 (a) Decide whether you think the following statement is true or false. If it is true, give a short explanation. If it is false, give a counterexample. [6]

"Let G be an arbitrary flow network, with a source s, a sink t, and a positive integer capacity C_e on every edge e; and let (A, B) be a minimum s-t cut with respect to these capacities $\{C_e : e \in E\}$. Now suppose we add 1 to every capacity; then (A, B) is still a minimum s-t cut with respect to these new capacities $\{1 + C_e : e \in E\}$ "

CO1 (b) Suppose you are coordinating the internship activities of the department. There are 'n' companies offering internships, and 'm' students have applied for the internship. The students choose more than one company for the internship based on their interest. Once choices from all the students are taken, your job is to assign internships to each student in a way that everyone is assigned exactly one company from the choices provided. [6]

Model this as a flow problem and describe a strategy to assign company to students based on their choice and also check if such an assignment is possible where every student gets a company of their choice?

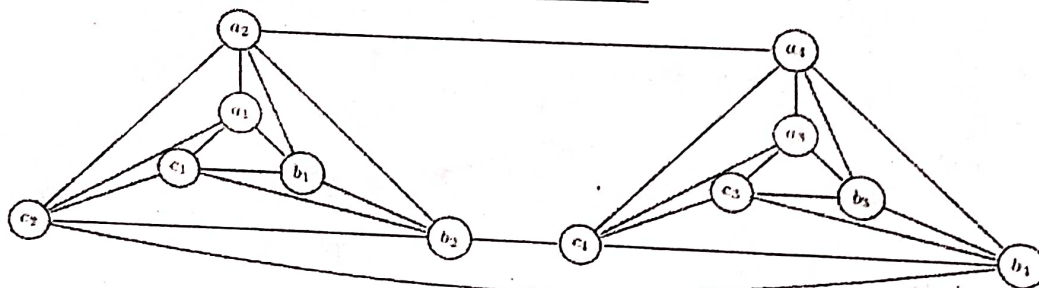
CO1 (c) Describe working of Ford Fulkerson's algorithm for the following s-t graph. [6]



Q.3 Attempt the following questions. [12]

CO2 (a) (i) Write and explain the Karger's algorithm to find the Min-Cut. Derive the probability of getting the Minimum Cut-set after $(n-2)$ Edge contractions. [6]

(ii) What is the size of the Min-Cut for the following graph G? The probability that the algorithm returns the Min cut is at least _____.



CO2 (b) Justify: Expected number of steps required to reach the Goal state (state n) from the starting state in the Random Walk for the 2-SAT problem is upper bounded by the square of the number of Boolean variables in the formula. [4]

CO1 (c) Let's say array A, which has 50 elements, is given to Randomized Quicksort algorithm: [2]

- What is the probability that the randomly selected pivot creates the worst partition?
- Given that the first partition was the worst, what is the probability that the 2nd partition is not the worst partition?

OR

CO2 (a) Write Fermat's Primality Testing algorithm. Assume $n=97$. Apply Randomized Primality Testing algorithm using Fermat's Theorem with randomly selected value of $a=29$. Check if it is F-Liar or F-Witness. Clearly show the steps of Computation. [6]

CO1 (b) What is the worst-case running time of Quicksort? How Randomization can be applied? Derive the upper bound on the Expected Number of Comparisons to sort n elements using the Randomized Quick sort. [4]

CO2 (c) Justify in brief: Randomized way to solve 3-SAT is not a polynomial time algorithm. [2]